

# Eric Ross *Senior/Staff Data Engineer*

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Versatile Staff Data Engineer with 10+ years of experience architecting and scaling data platforms across e-commerce and fintech environments. Proven ability to design and deploy cloud-native, analytics-ready systems using SQL, Python, dbt, Airflow, Snowflake, and Databricks. Specializes in building real-time pipelines, Medallion architecture, and automated observability frameworks. Partnered cross-functionally with ML, analytics, product, and engineering teams to drive business impact and enable predictive capabilities. Adept at mentoring engineers, embedding data contracts and governance practices, and improving end-to-end data reliability at scale.

## **PROFESSIONAL EXPERIENCE**

### **The Home Depot**

#### *Staff Data Engineer*

01/2021 – Present | Remote

- Led architecture of “StreamCore”, a streaming ingestion platform built on Kafka, AWS Glue, and Snowflake, processing 150M+ daily events for real-time personalization and monitoring.
- Developed the “SmartBundles Engine” with ML and pricing teams, leveraging Databricks, Delta Lake, and MLflow to generate bundled product recommendations, boosting average order value by 12%.
- Designed and maintained lakehouse infrastructure using Medallion architecture and auto-scaling Databricks clusters, improving data freshness SLAs and feature delivery.
- Created CI/CD pipelines using Terraform + GitHub Actions to automate dbt and Spark job deployments, versioning, and test execution.
- Built real-time fraud detection triggers with Kafka Streams + Snowflake Tasks, reducing false positives by 26% and speeding up incident resolution.
- Launched centralized data observability tools using Great Expectations and Prometheus, achieving 99%+ schema anomaly and freshness issue detection.
- Mentored 4+ engineers across data, ML, and platform teams while shaping engineering culture through an internal data engineering guild.

#### *Senior Data Engineer*

12/2015 – 12/2020 | Atlanta, GA

- Built the Unified Product Analytics Pipeline using Airflow, SQL, and Snowflake, streamlining SKU, inventory, and sales data for product analytics teams across e-commerce and retail.
- Migrated 50+ legacy Redshift jobs into modular dbt-based ELT workflows in Snowflake, cutting compute costs by 30% and increasing reliability.
- Partnered with analysts to model marketing attribution using dimensional modeling and SQL, enabling \$15M+ in revenue-impact dashboards.
- Designed and deployed A/B experiment pipelines using Firebase and Segment data, centralizing metrics in Snowflake for rapid analysis.
- Supported ML teams by developing shared batch features with Delta Lake, improving training efficiency and cross-team feature reusability.

### **CGI**

06/2013 – 12/2015 | Atlanta, GA

#### *Data Engineer*

- Developed secure customer transaction pipelines for Capital One using Python, SQL, Redshift, and OAuth2-authenticated data ingestion.
- Co-designed backend REST APIs and event tracking endpoints with mobile engineers to enable user behavior analytics via Amplitude and Firebase.
- Created clickstream enrichment pipelines for Macy’s e-commerce catalog using Java, AWS Lambda, and S3-based object versioning.
- Implemented automated data validation and unit testing for ETL jobs using Pytest and Pandas, improving deployment reliability by 40%.
- Collaborated with UX and analytics teams to create A/B test results pipelines, aggregating metrics via Segment + Snowflake for daily performance reviews.
- Contributed to client workshops on data lifecycle planning, metadata modeling, and cloud-native infrastructure strategy.

**SKILLS**

**Languages & Frameworks**

SQL, Python, PySpark, Java, Bash, dbt

**Cloud & Infra**

AWS (S3, Glue, Lambda, Redshift), Azure, Databricks, Terraform, GitHub Actions, Docker

**Modeling & Architecture**

Medallion Architecture, Star/Snowflake Schema, Data Vault, Dimensional Modeling

**Data Engineering & Orchestration**

Apache Airflow, dbt, Kafka, Spark, Delta Lake, Glue, Flink, Kinesis

**Data Platforms & Warehousing**

Snowflake, Databricks Lakehouse, Redshift, PostgreSQL, BigQuery

**Analytics & ML Tools**

Tableau, Looker, MLflow, Amplitude, Feature Store

**EDUCATION**

**University of Alabama at Birmingham**

*Bachelor's Degree in Computer Science*

09/2008 – 06/2013 | Birmingham, Alabama